GEK - 95334

DRIVE SYSTEM INSTRUCTIONS

EMERGENCY GENERATOR CONTROL PANEL

(WITH AUTOMATIC THROWOVER AND BACKFEEDING PROVISIONS)

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

EMERGENCY GENERATOR CONTROL PANEL

(WITH AUTOMATIC THROWOVER AND BACKFEEDING PROVISIONS)

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GEK - 95334

NOTICE

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, 1501 Roanoke Boulevard, Salem, Virginia, USA, 24153.

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SAFETY SYMBOL LEGEND

WARNING

Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

CAUTION

Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to, or destruction of equipment.

NOTE

Commands attention to an essential operating or maintenance procedure, condition or statement which must be highlighted.

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EMERGENCY GENERATOR CONTROL PANEL WITH AUTOMATIC THROWOVER/THROWBACK AND BACKFEED PROVISIONS

FUNCTION

The function of the Emergency Generator Control Panel is to provide rated power to critical loads on the emergency panel bus in the event of failure or shutdown (blackout) of the main supply generators.

OPERATION

Automatic and manual modes of operation are incorporated into the control, utilizing the features of electrically operated "Emergency Generator" and "Tie" circuit breakers. An automatic or manual mode operational selector switch is provided, in addition to a key lockable "Normal - Backfeed" selector switch. The positions of the switches will determine and govern the operational mode.

OPERATIONAL MODES

- Normal Mode
- Automatic Throwover Mode
- Automatic Throwback Mode
- Manual Mode
- Backfeed Mode

Normal Mode:

In the Normal mode of operation, the ship's service motor control center (MCC) provides power to the emergency generator power distribution panel via the "Tie" circuit breaker, which is closed. Power flows from the ship's service MCC through the "Tie" circuit breaker to critical loads on the emergency generator power distribution bus. The "Tie" circuit breaker is in "cascade" with a circuit breaker from the ship's service panel.

Automatic Throwover Mode

Should the normal flow of power via the "Tie" circuit breaker be interrupted, the "Tie" circuit breaker will trip on undervoltage.

Automatic Throwover Mode (Continued)

If the "Manual - Automatic" switch is in the Automatic Mode, a combination of this switch position coupled with tripping the "Tie" circuit breaker is permission to send a signal to the customer's emergency generator diesel engine, requesting engine start. Two items should be noted here, however, for they are critical to the automatic throwover mode.

- Cold start battery must be fully charged
- Engine control selector switch must be in LOAD position

During normal operation, the cold start battery circuit breaker must be left closed, so the main bus powered battery charger can keep it freshly charged. The cold start battery supplies the necessary governor power to start the engine and bring it to a self-sustaining idle speed (see Cold Start Battery Instructions).

The engine control selector switch must be in "LOAD" position to assure that the generator comes up to speed, automatically charges and closes the Emergency Generator circuit breaker, and supports the Emergency Generator Distribution Bus critical loads.

Automatic Throwback Mode

The "Automatic Throwback" mode of operation is used to return power to the normal mode of operation; that is, the ship's service motor control center supply.

In going back to the Normal mode of operation via the Automatic Throwback method, the following conditions are required:

- Emergency generator circuit breaker open
- Ship's service bus re-energized
- "Manual - Off - Auto" switch in AUTO position
- "Normal - Backfeed" switch in NORMAL position

When the above conditions exist, the "Tie" breaker receives control and under voltage power from the ship's service bus. The "Tie" breaker charge relay energizes instantaneously, causing the electrically operated "Tie" circuit breaker to charge, and also energizes the time delayed breaker closing relay. The breaker closing relay is delayed in pick-up (TDAE) to assure the breaker has had sufficient time to fully charge prior to receiving a close signal. (Normal charge time is 3.5 to 5.0 seconds.)

As control is returned to the Normal mode via the "Automatic Throwback" scheme, provisions are made to energize diesel engine "Cool Down" circuitry. The diesel engine "Cool Down" control circuit permits the diesel to operate unloaded (EGEN circuit breaker open) for a sufficient period of time to cool down prior to shutdown.

Manual Mode

In the Manual mode of operation, the "Manual - Off - Auto" selector switch is in the MANUAL position and all functions are operated manually. For example:

- Diesel engine is started manually, and
- Circuit breakers are charged and closed using manual pushbuttons.

In the Manual mode, all operations are controlled manually by the operator.

Backfeed Mode

The "Backfeed" mode is a manual operational mode which permits the emergency generator distribution bus to "Backfeed" the ship's service motor control center bus via the "Tie" circuit breaker in "cascade" with a circuit breaker off of the ship's service distribution bus.

The normal interlocking between the emergency generator circuit breaker and the "Tie" circuit breaker, which normally inhibits their simultaneous closing, is bypassed.

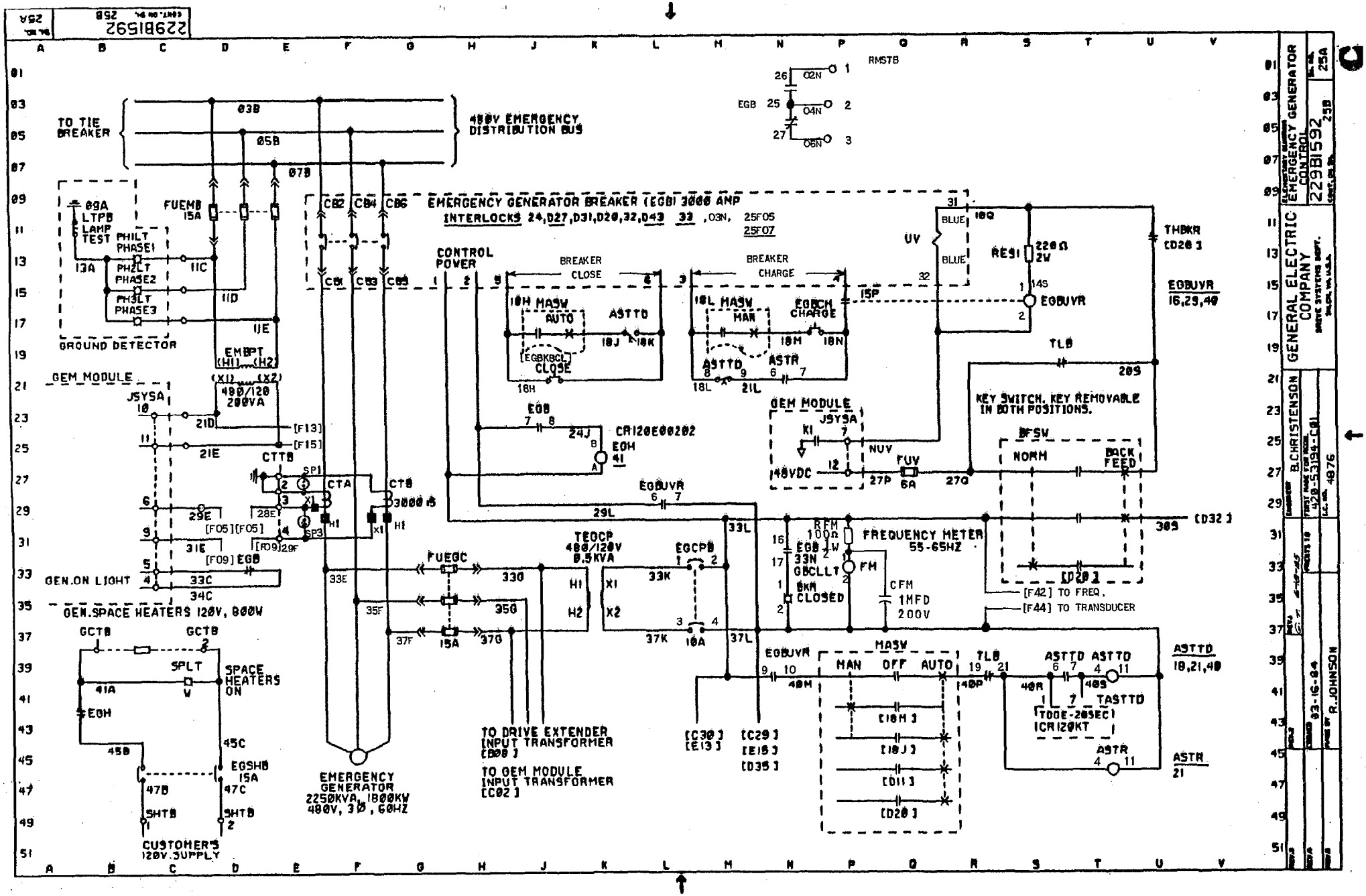
The "Normal - Backfeed" lockable control switch is placed in the "BACKFEED" position, which provides control and undervoltage power to the "Tie" circuit breaker. The "Tie" circuit breaker's "Automatic Throwback" mode is disabled at this time. This requires the operator to manually charge and close the "Tie" circuit breaker. Upon closing the "Tie" circuit breaker, the ship's service circuit breaker that is in "cascade" with the "Tie" circuit breaker may be closed. Moving the "Normal - Backfeed" switch back to NORMAL removes control and UV power from the "Tie" circuit breaker causing it to trip on undervoltage.

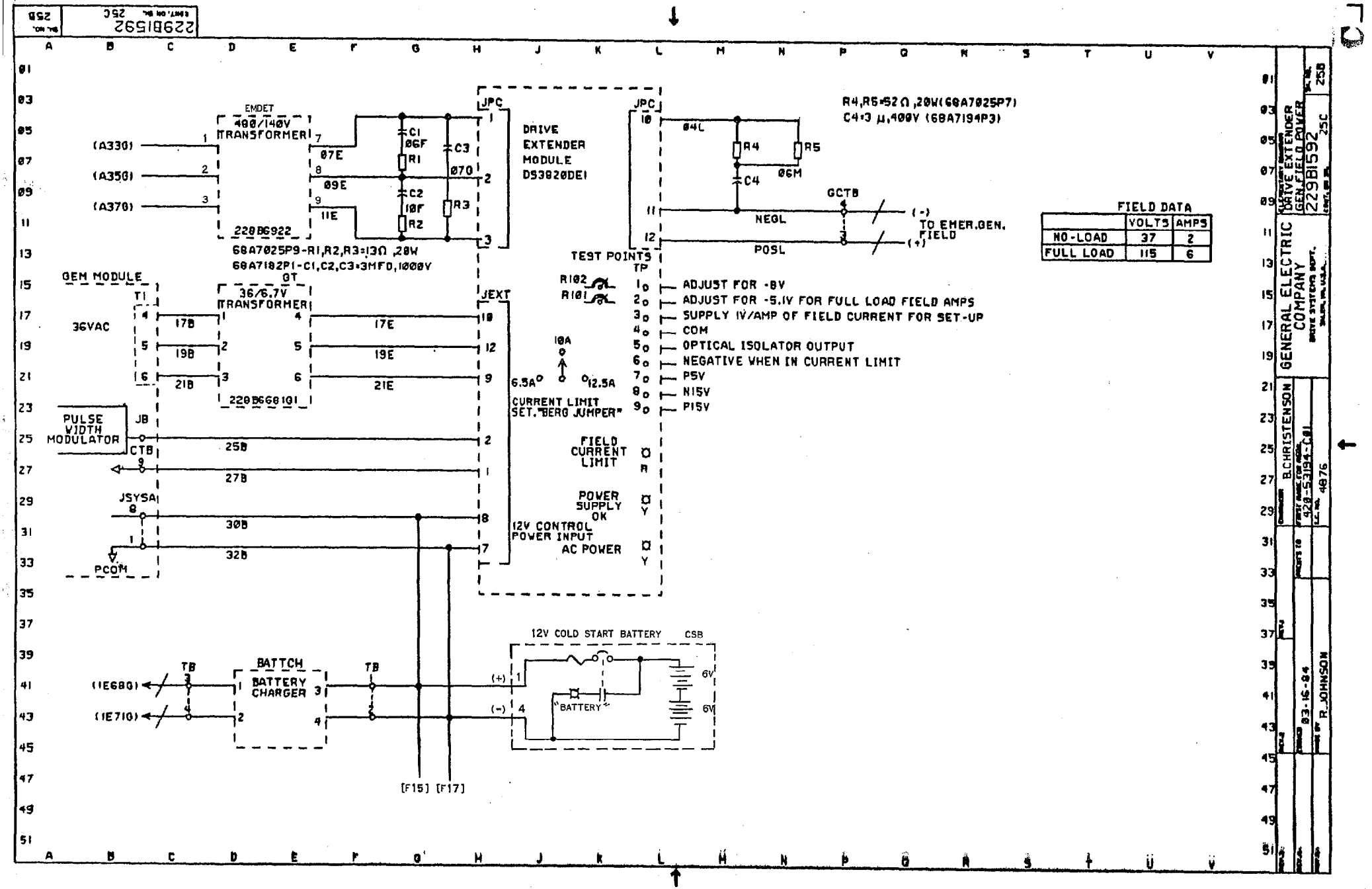
DRAWINGS AND DIAGRAMS

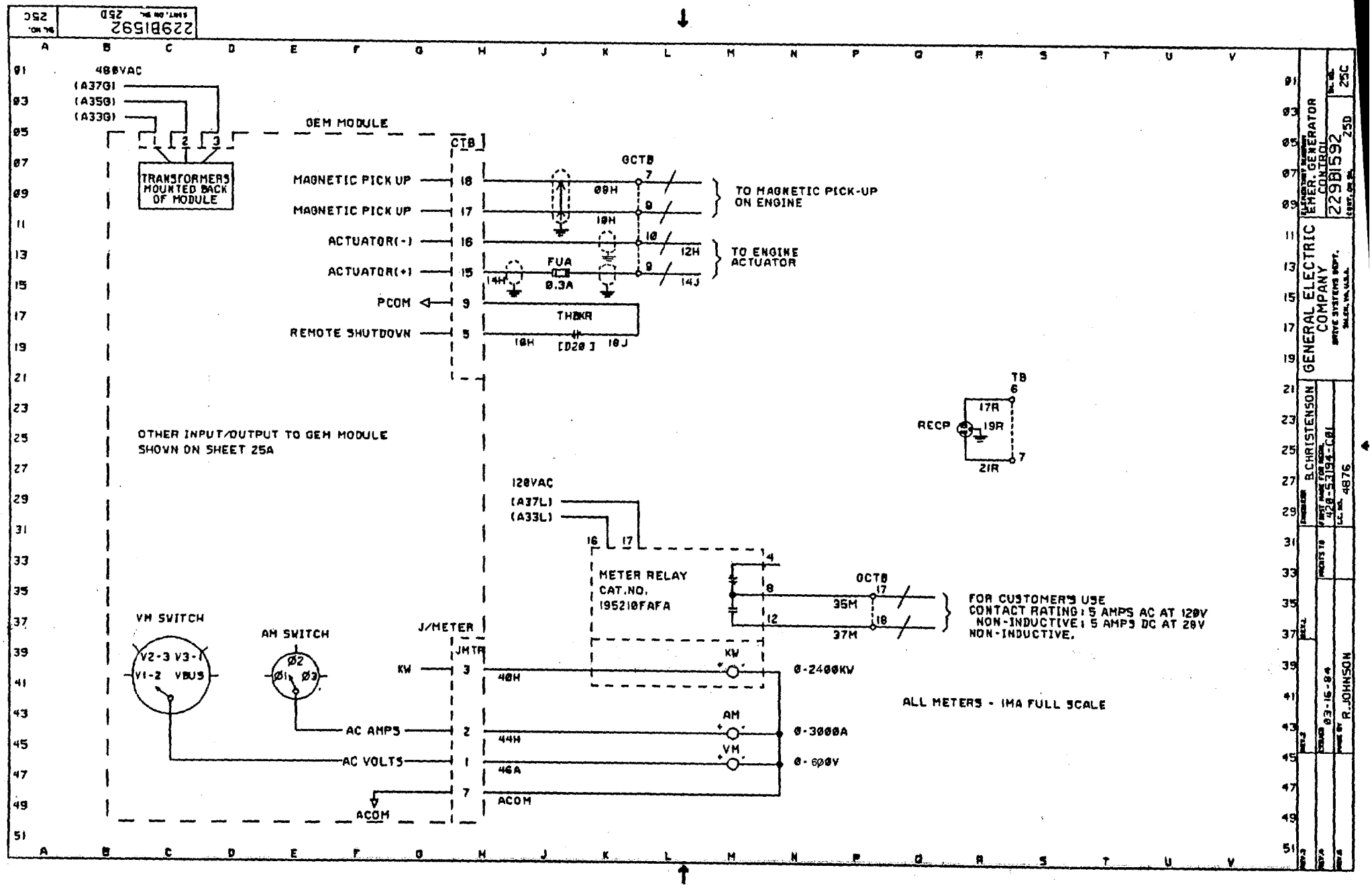
The remainder of this section contains drawings and diagrams that apply to the equipment.

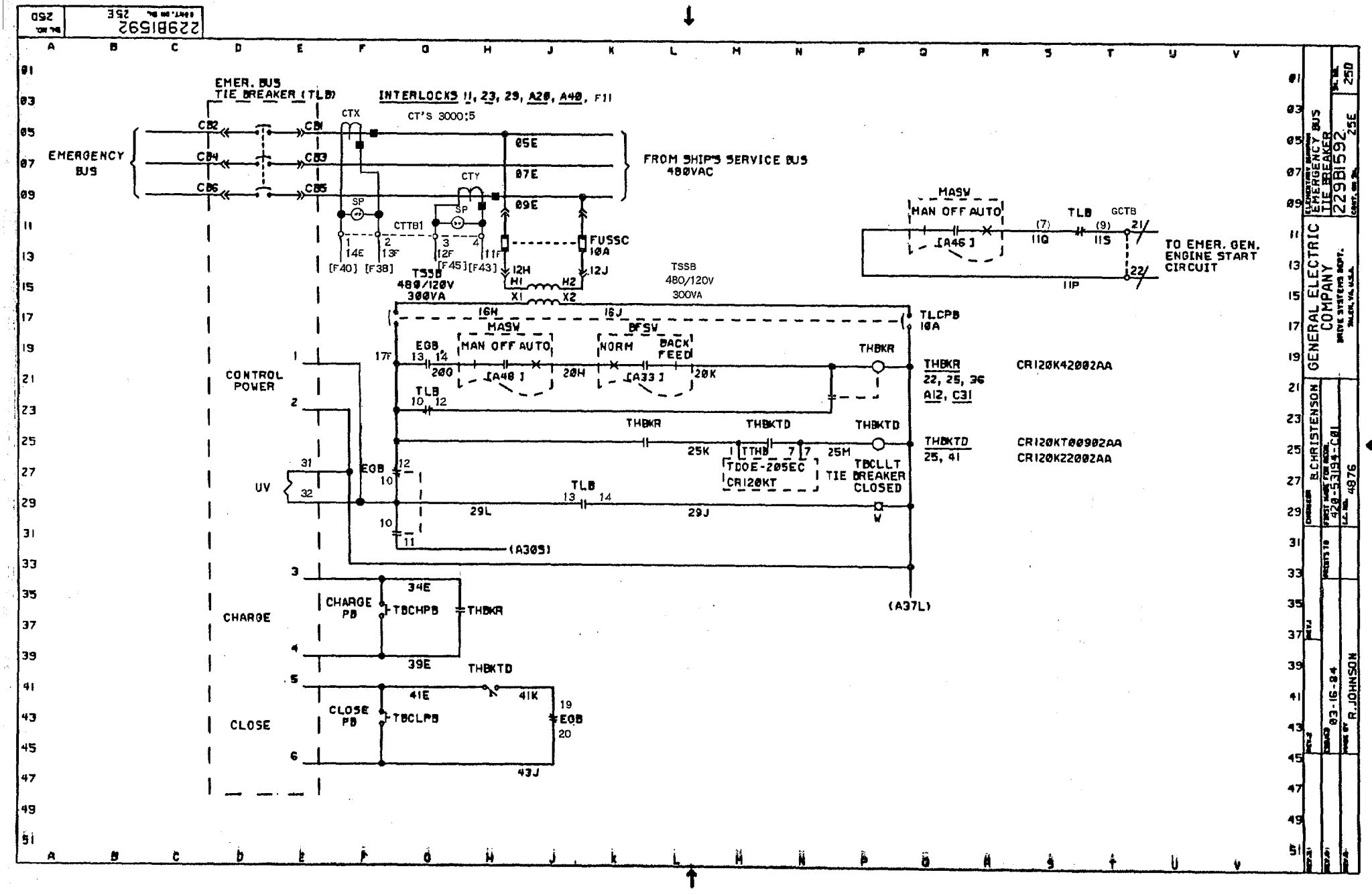
- Total System Diagram
- Emergency Distribution Control Circuits

NOTES

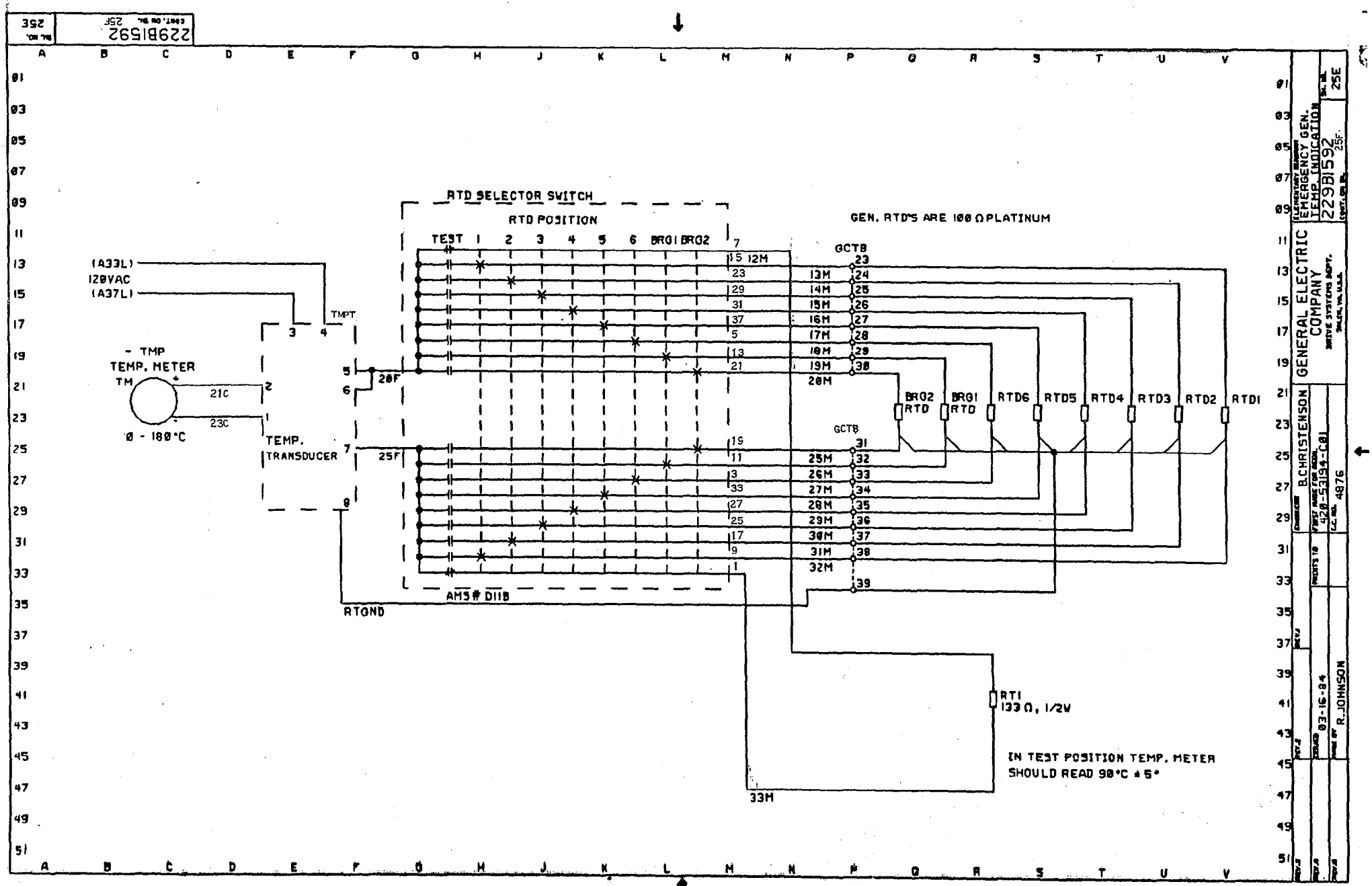

GENERAL ELECTRIC COMPANY
EMERGENCY GENERATOR CONTROL
229B1592
B.CHRISTENSON
TO TIE BREAKER
480V EMERGENCY DISTRIBUTION BUS
EMERGENCY GENERATOR BREAKER (EGB) 3000 AMP
INTERLOCKS 24,D27,D31,D20,32,D43 33 ,O3N, 25F05 25F07
GROUND DETECTOR
LAMP TEST
PH1LT PHASE1
PH2LT PHASE2
PH3LT PHASE3
GEM MODULE
480/120 200VA
CONTROL POWER
BREAKER CLOSE
BREAKER CHARGE
18H MASW
18L MASW
AUTO
MAN
ASTTD
ASTR
EGBCH CHARGE
[EGBKBCL] CLOSE
EGB
CR120E00202
EGH
41
EGBUVR
16,29,40
THBKR
[D28]
UV
RES1
220Ω 2W
14S
TLB
20S
KEY SWITCH, KEY REMOVABLE IN BOTH POSITIONS.
BFSW
NORM
BACK FEED
[D29]
30S
[D32]
[F42] TO FREQ.
[F44] TO TRANSDUCER
FREQUENCY METER
55-65HZ
RFM
100Ω
EGB
33N
GBCLLT
CLOSED
FM
CFM
1MFD
200V
480VDC
NUV
FUV
6A
CTTB
CTA
CTB
3000/5
TEGCP
480/120V
0.5KVA
EGCPB
FUEGC
15A
GEN.ON LIGHT
GEN.SPACE HEATERS 120V, 800W
GCTB
SPLT
SPACE HEATERS ON
EGSHB
SHTB
CUSTOMER'S 120V.SUPPLY
EMERGENCY GENERATOR 2250KVA, 1800KW 480V, 3Ø, 60HZ
TO DRIVE EXTENDER INPUT TRANSFORMER [B08]
TO GEM MODULE INPUT TRANSFORMER [C02]
[C30] [E13]
[C29] [E15] [D35]
MASW
MAN OFF AUTO
[18H] [18J] [D11] [D20]
TLB
ASTTD ASTTD
TASTTD
TDDE-20SEC
ICR120KT
ASTR
ASTTD 18,21,40
ASTR 21
RMSTB
EGB 25
02N
04N
06N
93-16-84
R.JOHNSON

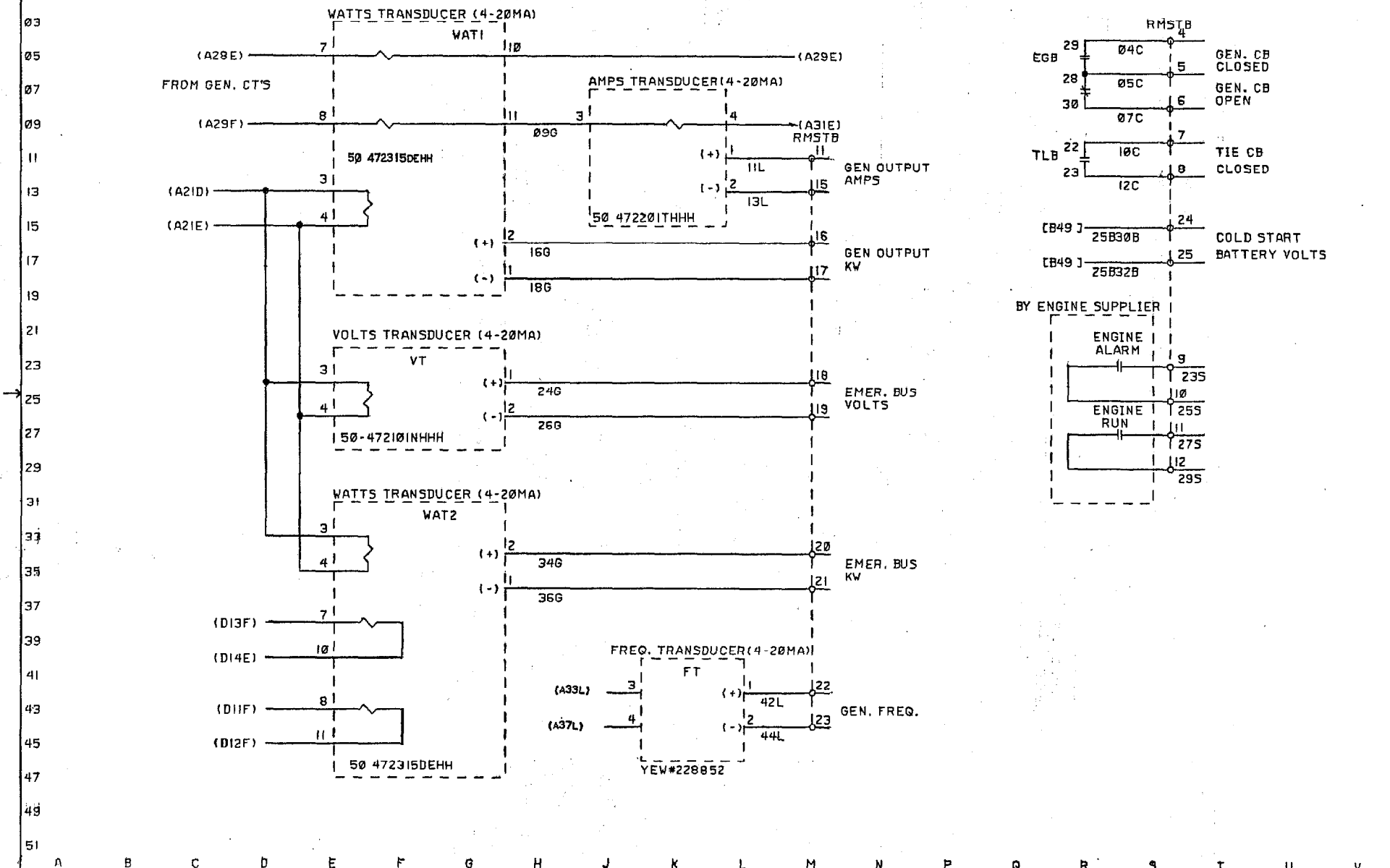

229B1592 25C
25B
DRIVE EXTENDER MODULE D93020DE1
JPC
JEXT
EMDET
400/140V TRANSFORMER
220B6922
68A7025P9-R1,R2,R3=13Ω ,20W
68A7182P1-C1,C2,C3=3MFD,1000V
GEM MODULE
T1
36VAC
GT
36/6.7V TRANSFORMER
220B6801
PULSE WIDTH MODULATOR
JB
CTB
JSYSA
PCOM
(A330)
(A350)
(A370)
C1 Ø6F
R1
C2 10F
R2
C3 Ø7D
R3
07E
09E
11E
17B
19B
21B
17E
19E
21E
25B
27B
30B
32B
R4,R5=52Ω ,20W(68A7025P7)
C4=3 μ,400V (68A7194P3)
R4
R5
C4
06M
04L
GCTB
NEGL
POSL
(-)
(+)
TO EMER.GEN. FIELD
TEST POINTS
TP
R102
R101
1 ADJUST FOR -8V
2 ADJUST FOR -5.1V FOR FULL LOAD FIELD AMPS
3 SUPPLY 1V/AMP OF FIELD CURRENT FOR SET-UP
4 COM
5 OPTICAL ISOLATOR OUTPUT
6 NEGATIVE WHEN IN CURRENT LIMIT
7 P5V
8 N15V
9 P15V
10A
6.5A
12.5A
CURRENT LIMIT SET."BERG JUMPER"
FIELD CURRENT LIMIT R
POWER SUPPLY OK Y
12V CONTROL POWER INPUT
AC POWER Y
FIELD DATA
VOLTS AMPS
NO-LOAD 37 2
FULL LOAD 115 6
12V COLD START BATTERY CSB
BATTCH
BATTERY CHARGER
TB
(1E680)
(1E710)
(+)
(-)
"BATTERY"
6V
6V
[F15] [F17]
GENERAL ELECTRIC COMPANY
DRIVE EXTENDER GEN.FIELD POWER
229B1592
25C
25B
B.CHRISTENSON
83-16-84
R.JOHNSON
4876


480VAC
(A37G)
(A35G)
(A33G)
GEM MODULE
TRANSFORMERS MOUNTED BACK OF MODULE
CTB
MAGNETIC PICK UP 18
MAGNETIC PICK UP 17
ACTUATOR(-) 16
ACTUATOR(+) 15
PCOM 9
REMOTE SHUTDOWN 5
GCTB
TO MAGNETIC PICK-UP ON ENGINE
TO ENGINE ACTUATOR
FUA 0.3A
THRKR
OTHER INPUT/OUTPUT TO GEM MODULE SHOWN ON SHEET 25A
TB
RECP
17R
19R
21R
120VAC
(A37L)
(A33L)
METER RELAY CAT.NO. 195210FAFA
GCTB
35M
37M
FOR CUSTOMER'S USE
CONTACT RATING: 5 AMPS AC AT 120V NON-INDUCTIVE; 5 AMPS DC AT 28V NON-INDUCTIVE.
VM SWITCH
V2-3 V3-1
V1-2 VBUS
AM SWITCH
Ø2
Ø1 Ø3
J/METER
JMTR
KW 3
AC AMPS 2
AC VOLTS 1
ACOM 7
KW
0-2400KW
AM
0-3000A
VM
0-600V
ALL METERS - 1MA FULL SCALE
GENERAL ELECTRIC COMPANY
EMER. GENERATOR CONTROL
22981592
25C
25D
B.CHRISTENSON
428-53194-C01
4876
03-16-84
R.JOHNSON



GENERAL ELECTRIC COMPANY
DRAWN B.CHRISTENSON
EMERGENCY BUS TIE BREAKER
229B1592
250
25E
03-16-84
R.JOHNSON
4876


RTD SELECTOR SWITCH
RTD POSITION
TEST 1 2 3 4 5 6 BRG1 BRG2 7
AMS# D11B
GEN. RTD'S ARE 100 Ω PLATINUM
GCTB
(A33L)
120VAC
(A37L)
TMPT
3 4
- TMP
TEMP. METER
TM
21C
23C
Ø - 180°C
TEMP. TRANSDUCER
2
1
5
6
20F
7
25F
8
RTGND
15 12M
23
29
31
37
5
13
21
19
11
3
33
27
25
17
9
1
13M
14M
15M
16M
17M
18M
19M
20M
25M
26M
27M
28M
29M
30M
31M
32M
33M
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
BRG2 RTD
BRG1 RTD
RTD6
RTD5
RTD4
RTD3
RTD2
RTD1
RTI
133 Ω, 1/2W
IN TEST POSITION TEMP. METER
SHOULD READ 98°C ± 5°
GENERAL ELECTRIC COMPANY
EMERGENCY GEN.
TEMP INDICATION
2298592
25E
25F
B.CHRISTENSON
420-53194-C01
4876
03-16-84
R.JOHNSON
2298592
25F
25E



GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
EMER. GEN. RIG MGMNT

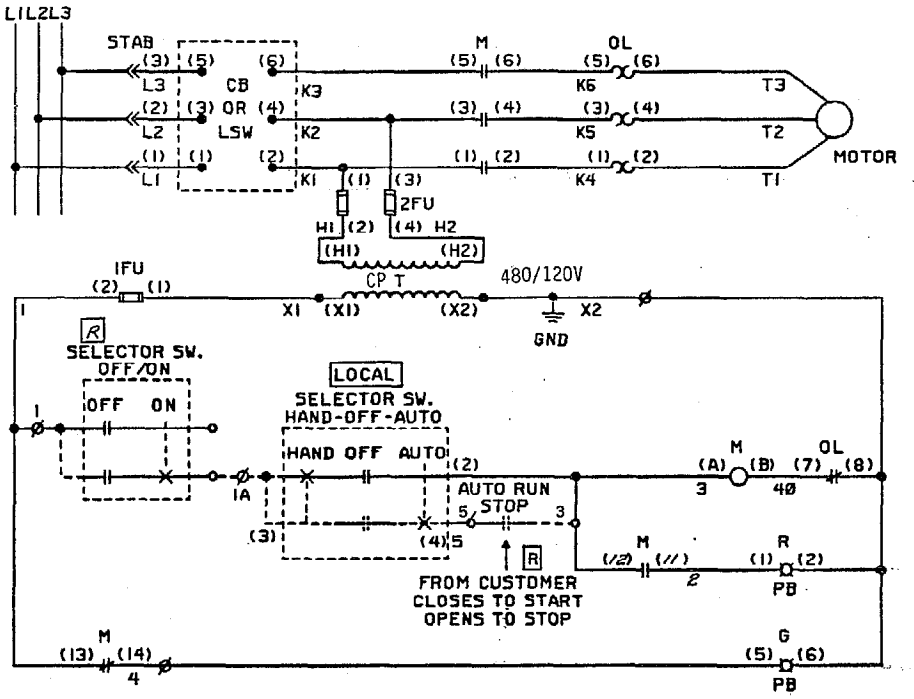
2298159Z 27A

SH. NO. 25F
CONT. ON SH.

ENGINEER B. CHRISTENSON
FIRST MADE FOR REGEN. 420-5319Y-C01
PRINTS TO
I.C. NO. 4876

REV.1
REV.2 ISSUED 06-04-84
REV.3
REV.4 MADE BY R. JOHNSON
REV.5

EMER. DIST. BUS
480V,3Ø,60HZ



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
EID	NO.2 BALLAST PRIMING PUMP-EP7	5	1	TECL Ø15	7.78A
KID	NO.1 BALLAST PRIMING PUMP-EP1	5	1	TECL Ø15	7.78A
FIF	NO.3 BALLAST PRIMING PUMP-EP62	5	1	TECL Ø15	7.78A
FIH	NO.4 BALLAST PRIMING PUMP-EP63	5	1	TECL Ø15	7.78A

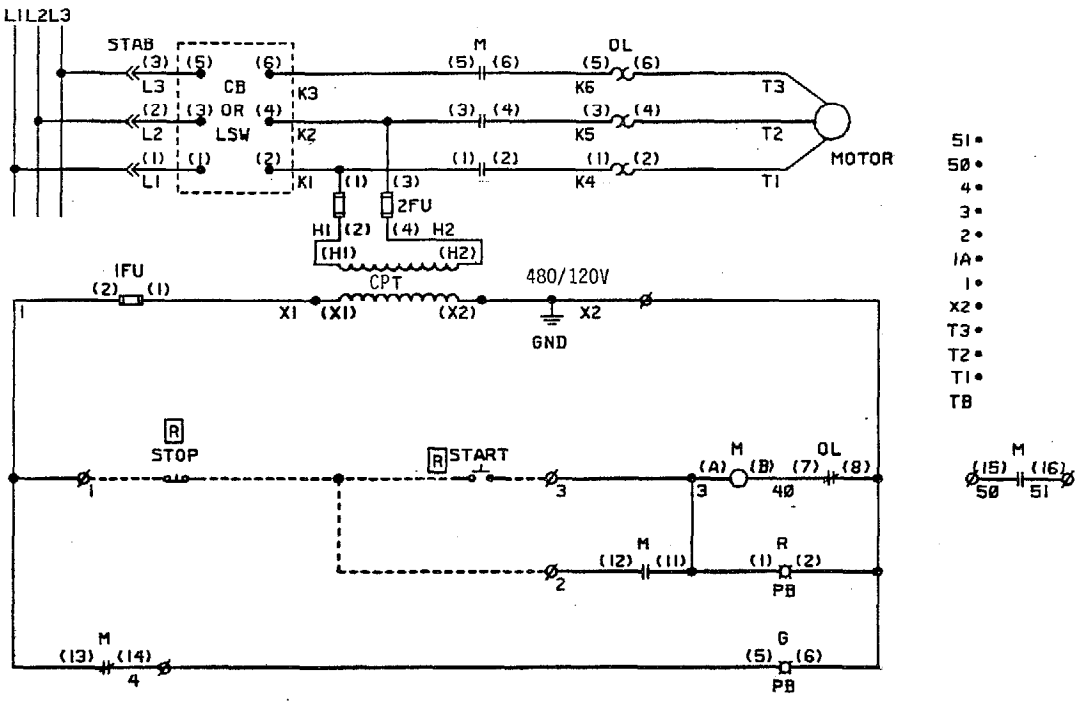
CONTROL CIRCUIT DESCRIPTION:

- REMOTE ON/OFF SELECTOR SWITCH
- LOCAL HAND/OFF/AUTO SELECTOR SWITCH
- REMOTE AUTO RUN/STOP CONTACT PROVIDED BY CUSTOMER (HOA MUST BE IN AUTO TO FUNCTION PROPERLY)
- LVR - LOW VOLTAGE RETURN - PERMITS RESTART OF UNIT TRIPPED OUT ON LOW VOLTAGE AUTOMATICALLY.

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

-A-2HC21

REV.3	REV.2	REV.1	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQN. 420-53194-C01	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	EMER. DIST. STARTERS
REV.5	MADE BY R. JOHNSON		I.C. NO.		229B1592 28C CONT. ON SH. / SH. NO. 28B



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
HID	SPARE STARTER NO.6 - EP13	15	2	TECL 30	21.4B
HIN	CEMENT MIX PUMP - EP14	50	3	TECL 100	77.2B
JIN	SPARE STARTER NO.3 - EP15	50	3	TECL 100	77.2B
NIF	SPARE STARTER NO.2 - EP12	25	2	TECL 50	33.ØB
KIN	SPARE STARTER NO.4 - EP17	50	4	TECL 100	77.2B

CONTROL CIRCUIT DESCRIPTION:

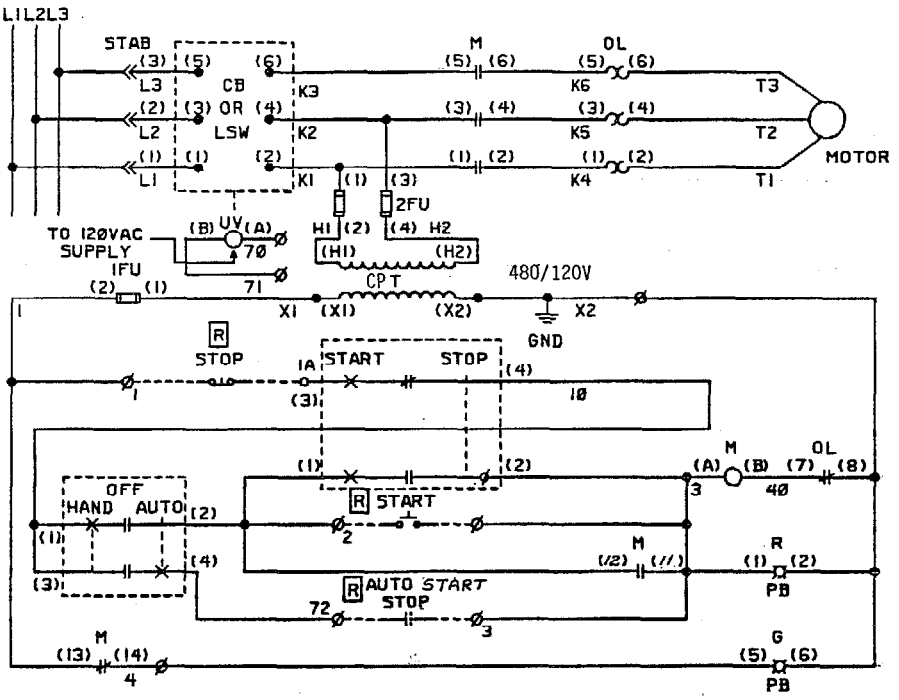
- REMOTE START/STOP PUSH BUTTONS PROVIDED BY CUSTOMER
- LVP - LOW VOLTAGE PROTECTION - PREVENTS RESTART OF UNIT AUTOMATICALLY ON LOW VOLTAGE LOSS/RETURN.

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

-A-2HC21

REV.3	REV.2	REV.1	ENGINEER	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM EMER. DIST. STARTERS
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQN. 420-53194-C01		229B1592 CONT. ON SH. 28D / SH. NO. 28C
REV.5	MADE BY R. JOHNSON		I.G. NO.		

EMER. DIST. BUS
480V,3Ø,60HZ



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
CID	NO.1 FUEL OIL TRANSFER PUMP-EP2	15	2	TECL 30	21.4B

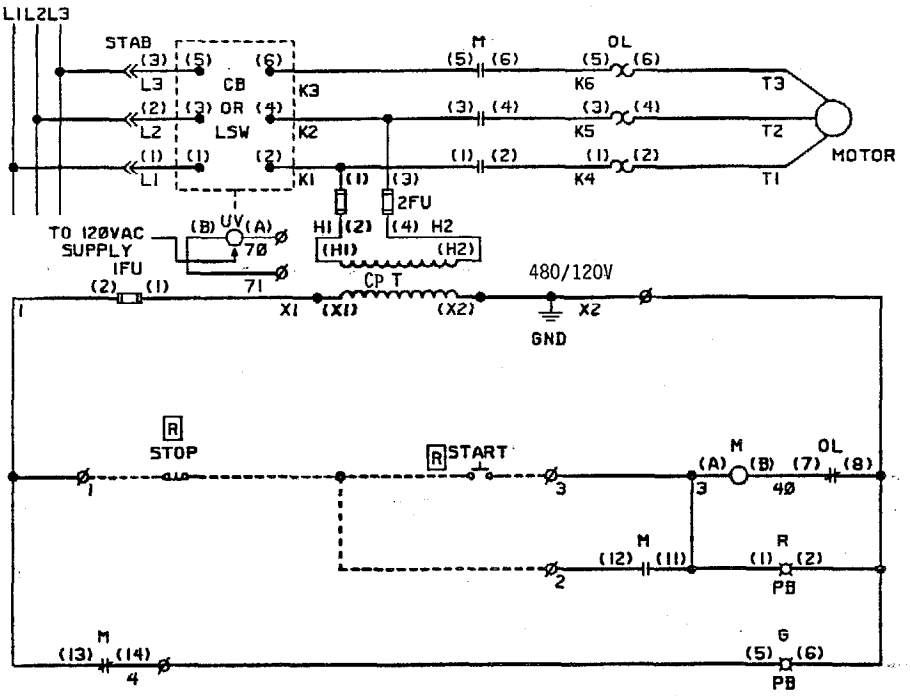
CONTROL CIRCUIT DESCRIPTION:

- SAME AS CKT. ON SH. 28A, EXCEPT UV DEVICE ON CB. (CB MUST BE MANUALLY RESET (LOCALLY) PRIOR TO ACTIVATION OF ANY PILOT DEVICE)
- LOCAL START/STOP SELECTOR SWITCH SPRING RETURNS TO NEUTRAL POSITION.
- LVP - LOW VOLTAGE PROTECTION - PREVENTS RESTART OF STARTER UNIT AUTOMATICALLY ON LOW VOLTAGE LOSS/RETURN.

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

-A-2HC21

REV.3	REV.2	REV.1	ENGINEER AWA	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM EMER. DIST. STARTERS
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQM. 420-53194-C01		229B1592 28E / SH. NO. 28D
REV.5	MADE BY R. JOHNSON		I.C. NO.		CONT. ON SH.



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
DID	FUEL OIL - EP5	7.5	1	TECL Ø15	12.5B
JID	SPARE STARTER NO.5 - EP3	15	2	TECL Ø30	21.4B
LID	SPARE STARTER NO.1 - EP6	7.5	1	TECL Ø15	12.5B

CONTROL CIRCUIT DESCRIPTION:

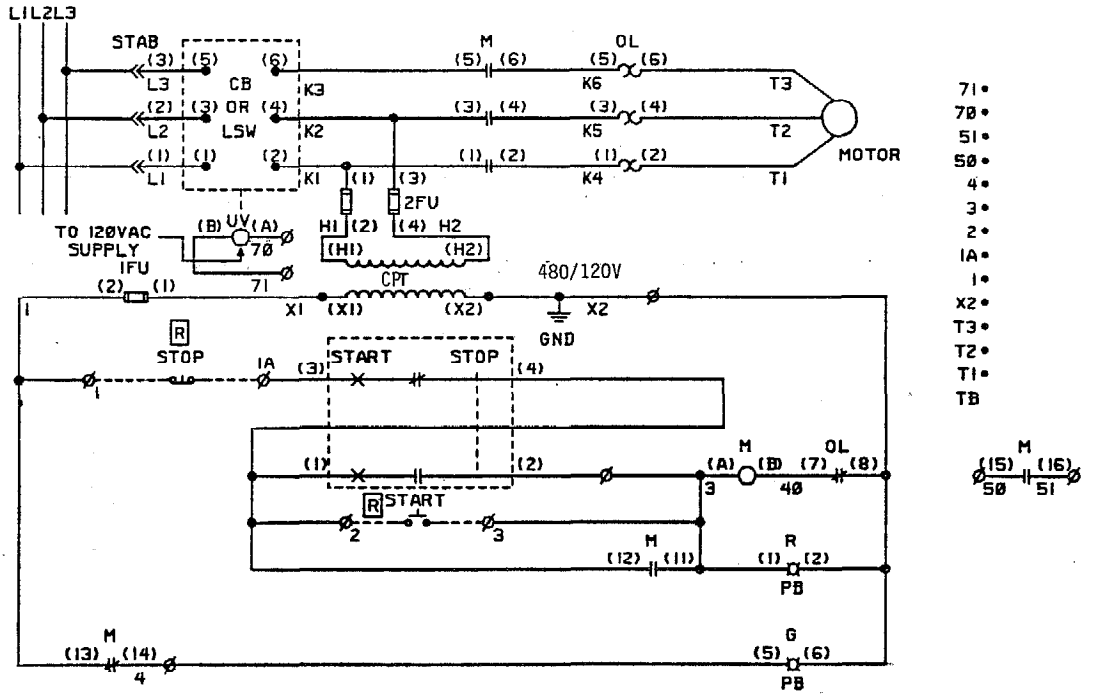
*SAME AS CIRCUIT ON SHEET 28C, EXCEPT CB MUST BE LOCALLY RESET ON UNDERVOLTAGE TRIP.
*LVP - LOW VOLTAGE PROTECTION.

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

-A-2HC21

REV.3	REV.2	REV.1	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQN. 420-53194-C01	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	EMER. DIST. STARTERS
REV.5	MADE BY R. JOHNSON		I.C. NO.		229B1592 CONT. ON SH. 28F / SH. NO. 28E

C



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
BID	EMERG. GEN ROOM EXHAUST FAN-EP4	2	1	TECL Ø07	3.56A

CONTROL CIRCUIT DESCRIPTION:

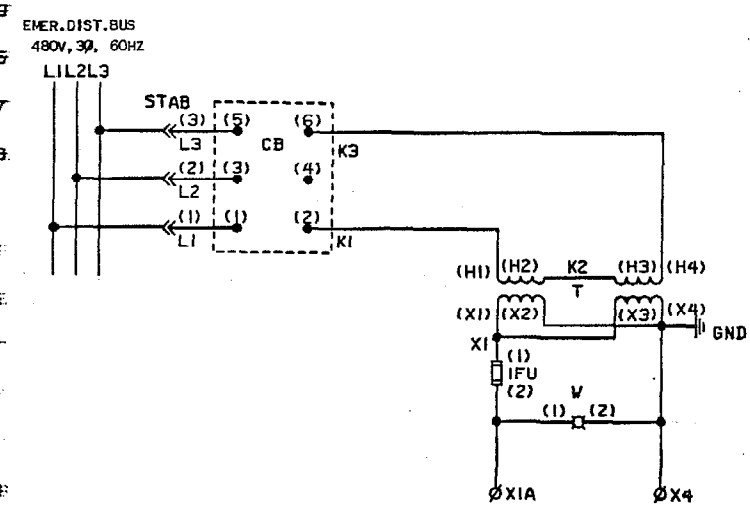
- LOCAL START/STOP SELECTOR SWITCH MUST BE HELD IN START POSITION LONG ENOUGH TO ENERGIZE AND SEAL IN "M" CONTACTOR. SPRING RETURN TO NEUTRAL POSITION.
- REMOTE START/STOP PUSH BUTTONS.
- CIRCUIT BREAKER MUST BE MANUALLY & LOCALLY RESET AFTER AN UNDERVOLTAGE CONDITION.
- LVP - LOW VOLTAGE PROTECTION.

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

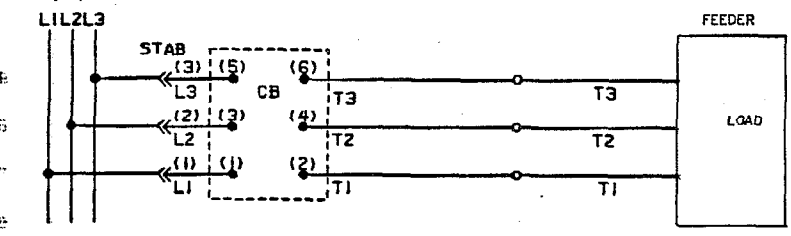
-A-2HC21

REV.3	REV.2	REV.1	ENGINEER	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM EMER. DIST. STARTERS
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REGN. 420-53194-C01		229BI592 CONT. ON SH. 28G SH. NO. 28F
REV.5	MADE BY R. JOHNSON		I.O. NO.		

C



UNIT	DESIGNATION	DISCONNECT
GIF	PORT UV POWER	TEDLØ15
JIJ	STBD UV POWER	TEDLØ15

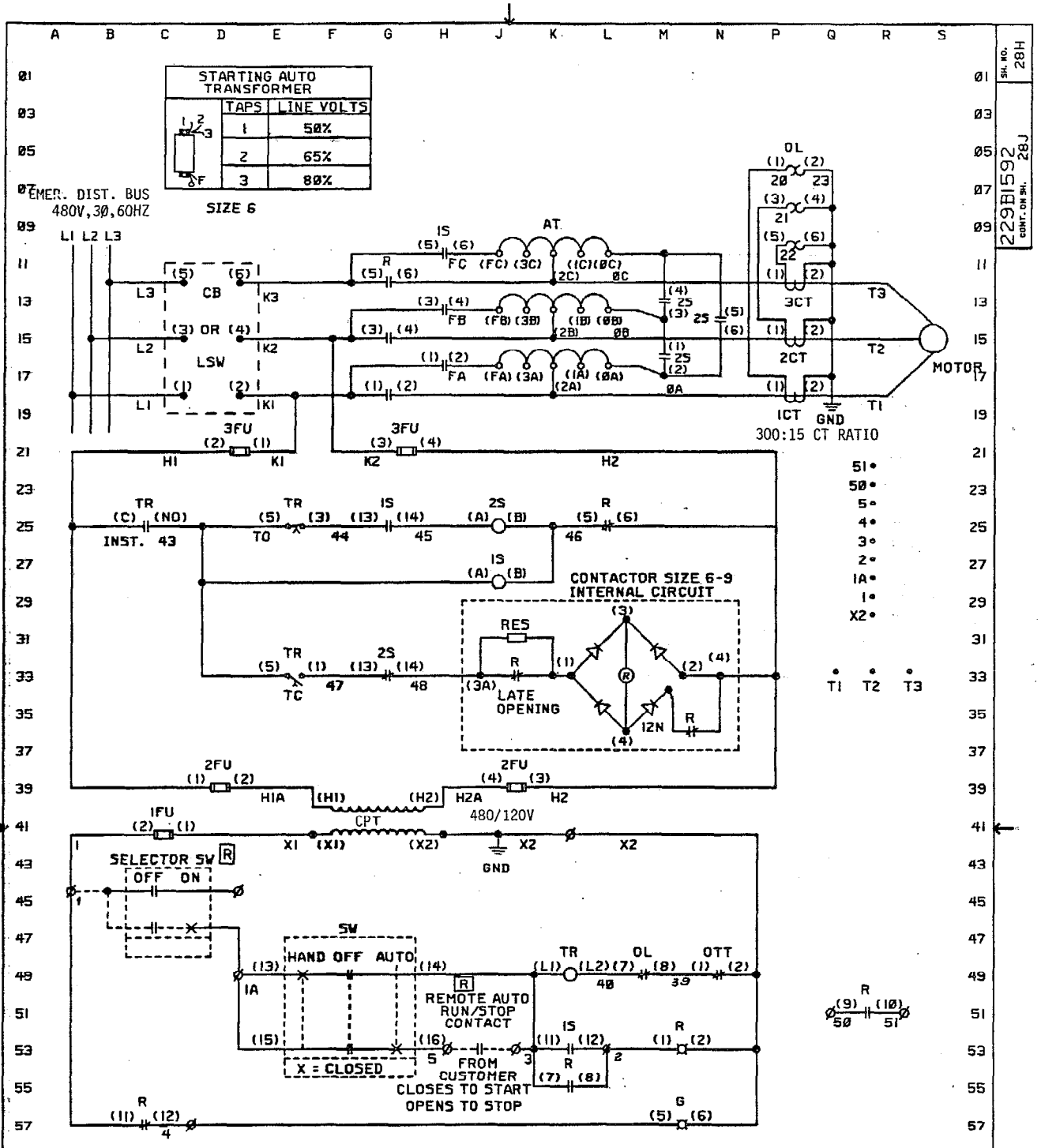


UNIT	DESIGNATION	DISCONNECT
AIE	VALVE HYD PUMP #1-EP35	TEDL Ø30
AIF	UPS PANEL 20KVA EP48	TEDL Ø80
AIN	NO.1 REG SERVICE AIR COMP. EP23	TBC4 400
BIE	NO.1 ELEVATOR-EP25	TEDL Ø40
BIF	NO.3 ELEVATOR-EP27	TEDL Ø40
BIG	NO.1 BOP HYDC CONTROL UNIT-EP29	TEDL Ø80
CIE	WT DOOR DISTR. PANEL STDB SIDE - EP31	TEDL Ø20
CIF	37.5KVA RMS TRANS.-EP57	TEDL 100
CIG	EMGR GEN BATTERY CHARGER-EP45	TEDL 015
DIE	SPARE FEEDER NO.10-EP43	TEDL Ø30
DIF	FUEL OIL FEEDPUMP MCC-EP34	TEDL Ø20
EIE	SPARE FEEDER NO.4 - EP33	TEDL 100
EIF	SPARE FEEDER NO.3 - EP49	TEDL 015
EIG	SPARE FEEDER NO.11 - EP42	TEDL 050
FII	SPARE FEEDER NO.12 - EP53	TEDL 100
FIN	NO.1 EMERGENCY TRANSFORMER-EP58	TP4 225
GIN	SPARE FEEDER #14 - EP61	TB4 200
HIF	DIVING UNIT-EP55	TBI 125
HIG	SPARE FEEDER NO.9 - EP44	TEDL Ø20
HII	UPS PANEL 30KVA-EP52	TBI 125

CIRCUIT DESCRIPTIONS
GIF/JIJ - UNDER VOLTAGE DEVICE POWER SOURCE._____ VA, 10.
REMAINDER - FEEDER UNITS

TIAA5B

REV.3	REV.2	REV.1	ENGINEER RWH	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM	
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQN. 420-53194-C01		229B1592 CONT. ON SH. 28H	SH. NO. 28G
REV.5	MADE BY R. JOHNSON		I.C. NO.			



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
PIN	NO.2 FIRE PUMP AND WASHDOWN PUMP - EP24	250	6		3.01A
QIN	NO.2 FIRE PUMP AND WASHDOWN PUMP - EP24	DISCONNECT FOR UNIT PIN	TBC6	600	

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

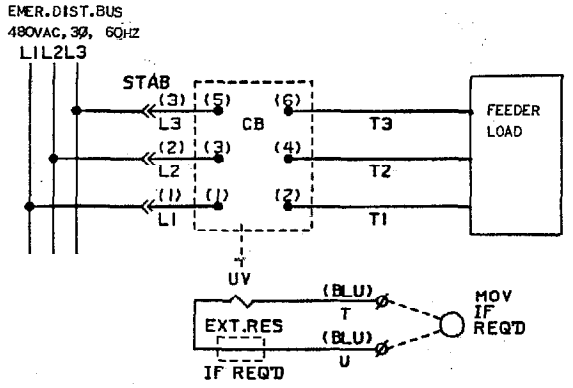
CONTROL CIRCUIT DESCRIPTION:

- SAME AS CKT ON SH. 28B, IN ADDITION TO CLOSED TRANSITION AND TIMING FOR REDUCED VOLTAGE NON REV. (RVNR) AUTO TRANSFORMER STARTING
- LVR - LOW VOLTAGE RETURN.

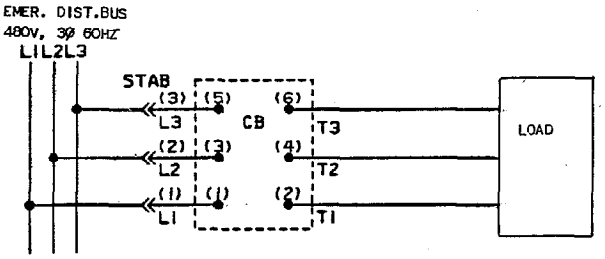
6A-2HC21

REV.3	REV.2	REV.1	ENGINEER A.W.B	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM RVNR AUTO XFMR. NEMA 6 STARTER
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQN. 420-53194-C01		229B1592 CONT. ON SH. 28J / SH. NO. 28H
REV.5	MADE BY R. JOHNSON		I.C. NO.		

SH. NO. 28H
229B1592 CONT. ON SH. 28J



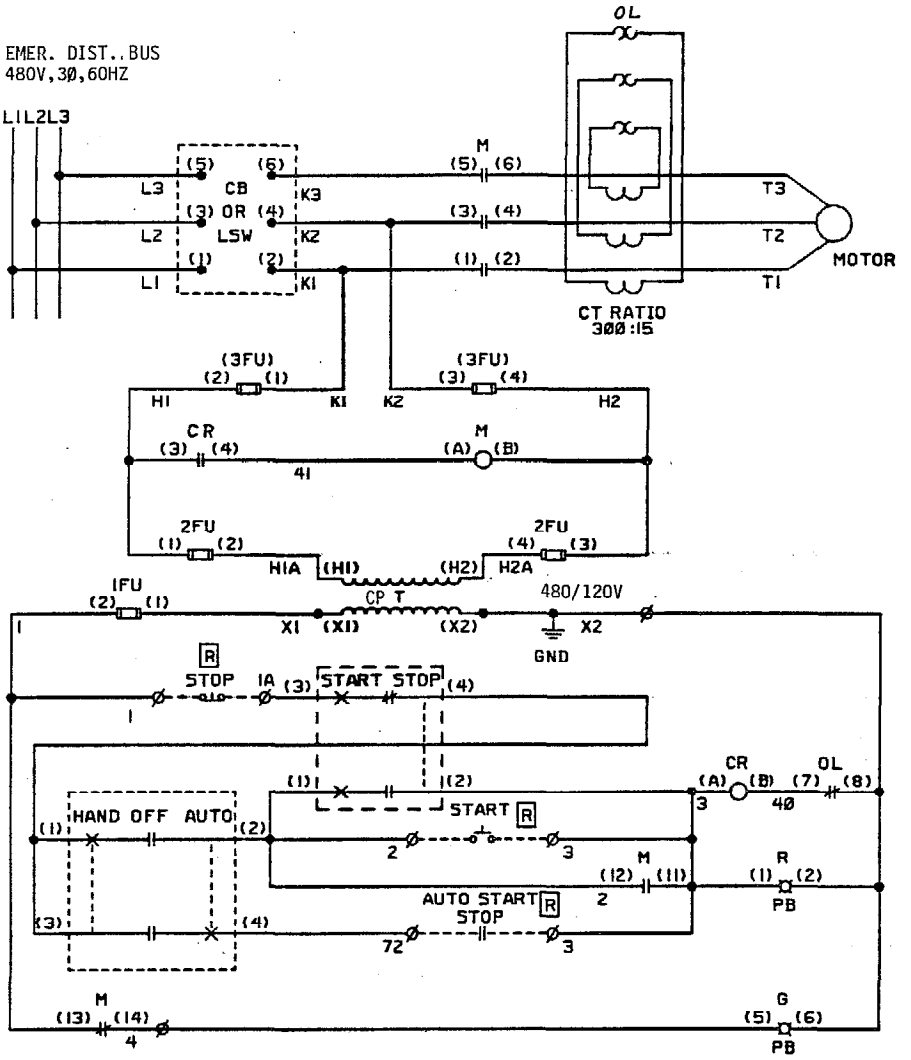
UNIT	DESIGNATION	DISCONNECT	
MIF	STEAM GEN-EP56	TBI	150
JIH	EMGR GEN ROOM SUPPLY FAN-EP38	TEDL	050
AIG	SPARE FEEDER #2-EP40	TEDL	015



UNIT	DESIGNATION	DISCONNECT	
JIE	CEMENT PUMP ROOM FAN-EP59	TEDL	015
JIF	25KVA RMS TRANSF.- EP51	TEDL	070
JIG	SPARE FEEDER NO.13 EP54	TEDL	025
KIE	WT DOOR DISTR.PANEL PORT SIDE-EP32	TEDL	020
KIF	SPARE FEEDER NO.6-EP37	TEDL	100
KIG	SPARE FEEDER NO.7-EP50	TEDL	015
KIH	ELEVATOR NO.4-EP28	TEDL	040
KII	SPARE FEEDER NO.8-EP41	TEDL	050
LIE	ELEVATOR NO.2-EP26	TEDL	040
LIF	SPARE FEEDER NO.1	TEDL	080
LIG	DIVERTER HOUSING EXH.FAN-EP39	TEDL	015
LIH	SPARE FEEDER NO.5-EP36	TEDL	100
LII	FUEL OIL SERVICE PUMP-EP46	TEDL	015
LIN	NO.2 EMERGENCY TRANSFORMER-EP60	TB4	225
NIG	EMERGENCY GEN.FUEL OIL FEED PUMP-EP47	TEDL	015

FBI-00U

REV.3	REV.2	REV.1	ENGINEER A.W.H	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM FEEDERS UV AND NON UV.
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQN. 420-53194-C01		229B1592 CONT. ON SH. 28K SH. NO. 28J
REV.5	MADE BY R. JOHNSON		I.C. NO.		C



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
BIN	NO.3 BALLAST PUMP-EP20	150	5	TBC4 400	9.55A
MIN	NO.1 BALLAST PUMP-EP19	150	5	TBC4 400	9.55A
NIN	NO.2 BALLAST PUMP-EP21	150	5	TBC4 400	9.55A

CONTROL CIRCUIT DESCRIPTION:

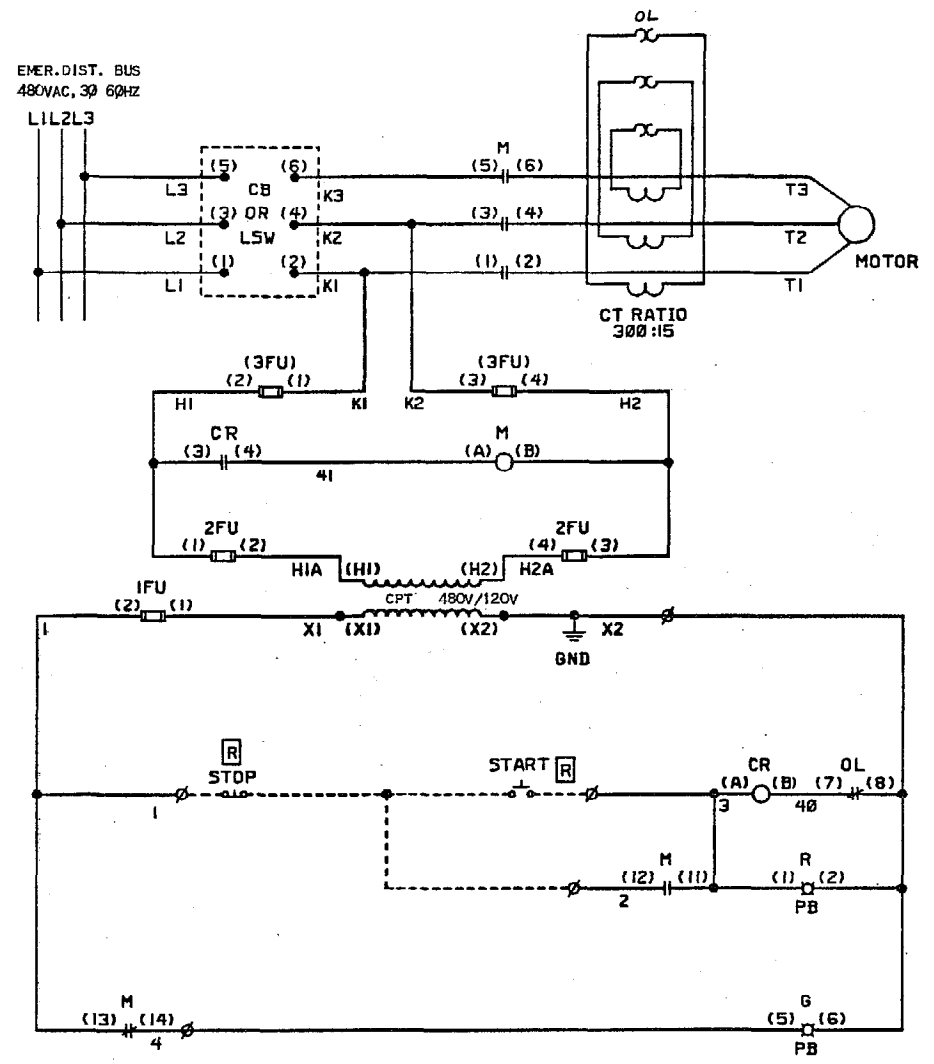
SAME AS CIRCUIT ON SH. 28A, HOWEVER STARTERS ARE NEMA SIZE 5

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

5A-2HC21

REV.3	REV.2	REV.1	ENGINEER AWA	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM PUMPS STARTERS BALLAST
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQM. 420-53194-C01		229B1592 CONT. ON SH. 28L
REV.5	MADE BY R. JOHNSON		I.C. NO.		SH. NO. 28K

C



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
DIN	NO.1 MUD MIX PUMP-EP18	125	5	TBC4 400	77.88

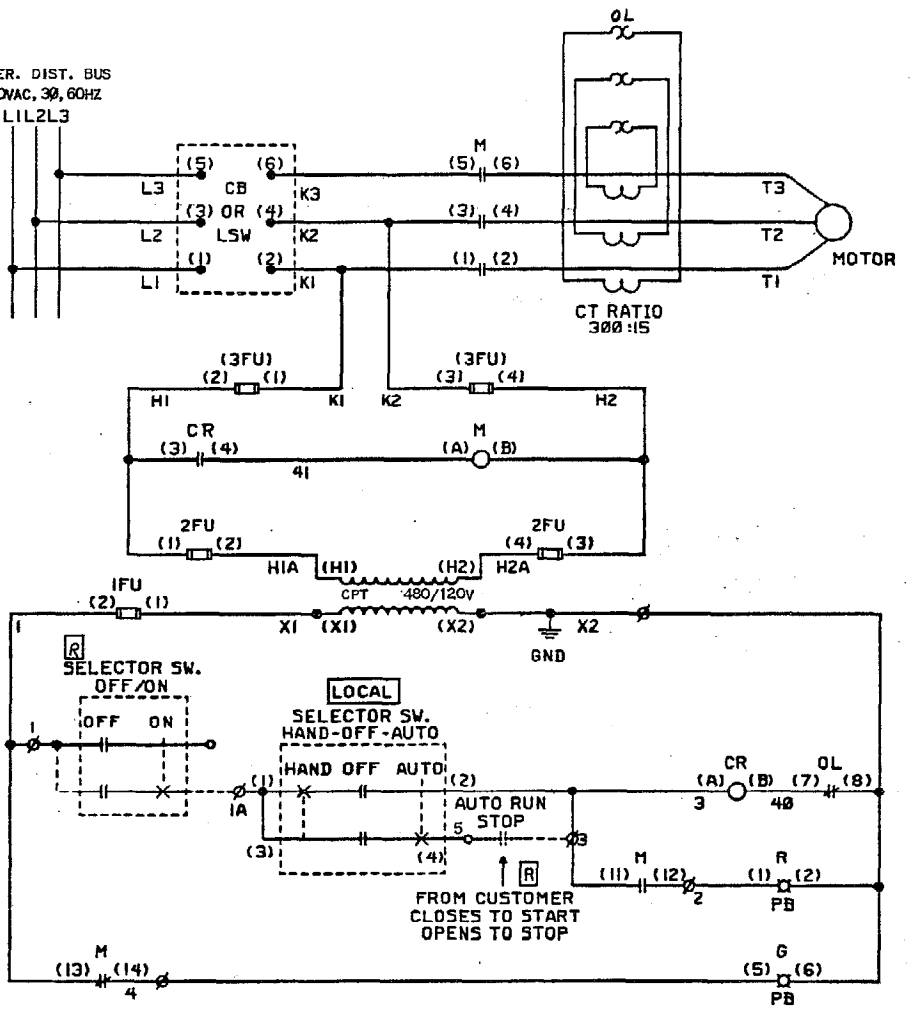
CONTROL CIRCUIT DESCRIPTION

- SAME AS SHOWN ON SH. 28C, HOWEVER STARTER IS NEMA SIZE 5
- LVP -LOW VOLTAGE PROTECTION.

[R] REMOTE DEVICE, SUPPLIED BY CUSTOMER

5A-2HC21

REV.3	REV.2	REV.1	ENGINEER R.W.H.	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM MUD MIX PUMP NO. 1
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REQN. 420-53194-C01		229B1592 CONT. ON SH. 28M SH. NO. 28L
REV.5	MADE BY R. JOHNSON		I.C. NO.		C



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
CIN	NO.4 BALLAST PUMP-EP22	150	5	TBC4 400	9.55A

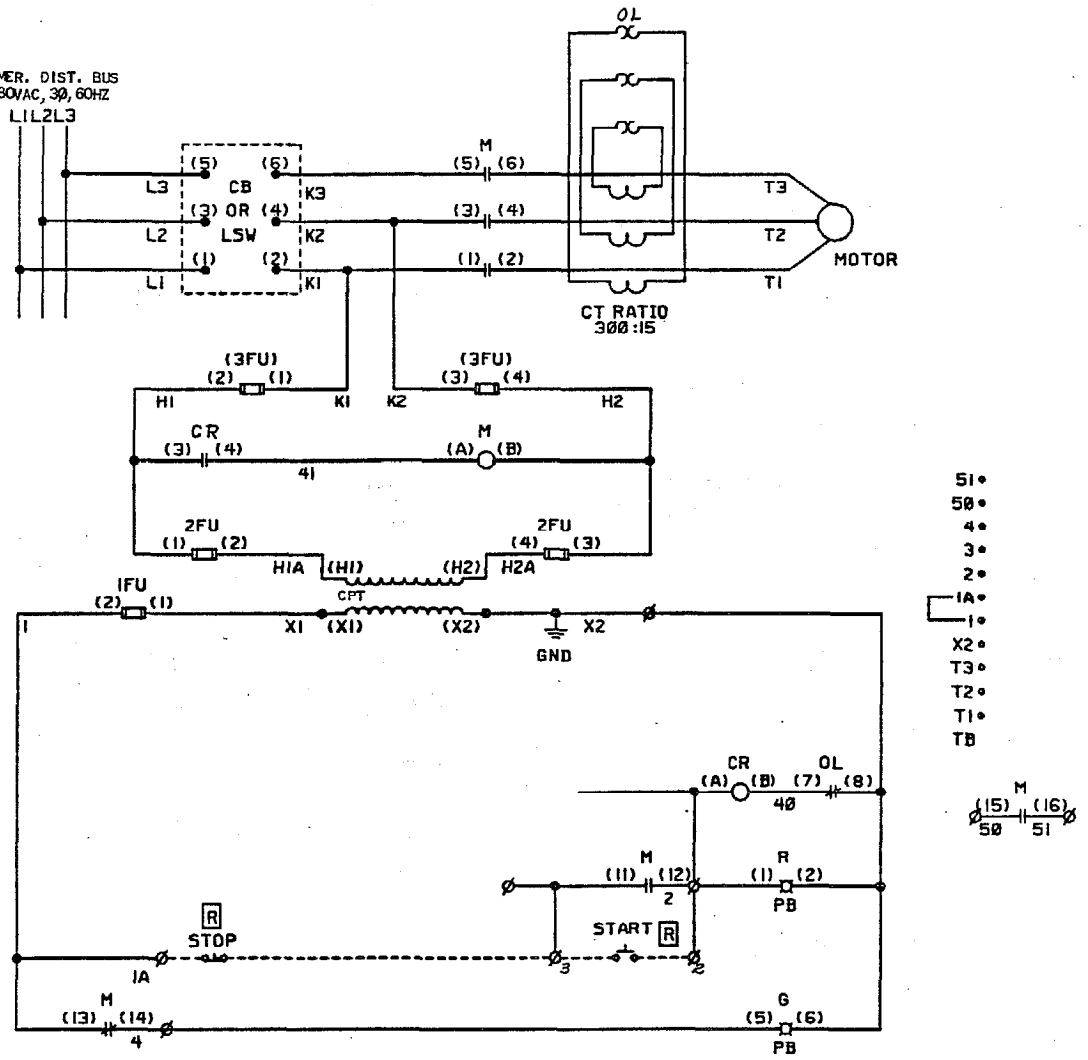
CONTROL CIRCUIT DESCRIPTION:

•SAME AS SHOWN ON SH. 28B

R REMOTE DEVICE, SUPPLIED BY CUSTOMER

5A-2HC21

REV.3	REV.2	REV.1	ENGINEER AWD	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM NO. 4 BALLAST PUMP - EP22
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REGN. 420-53194-C01		229B1592 CONT. ON SH. 28N — SH. NO. 28M
REV.5	MADE BY R. JOHNSON		I.C. NO.		



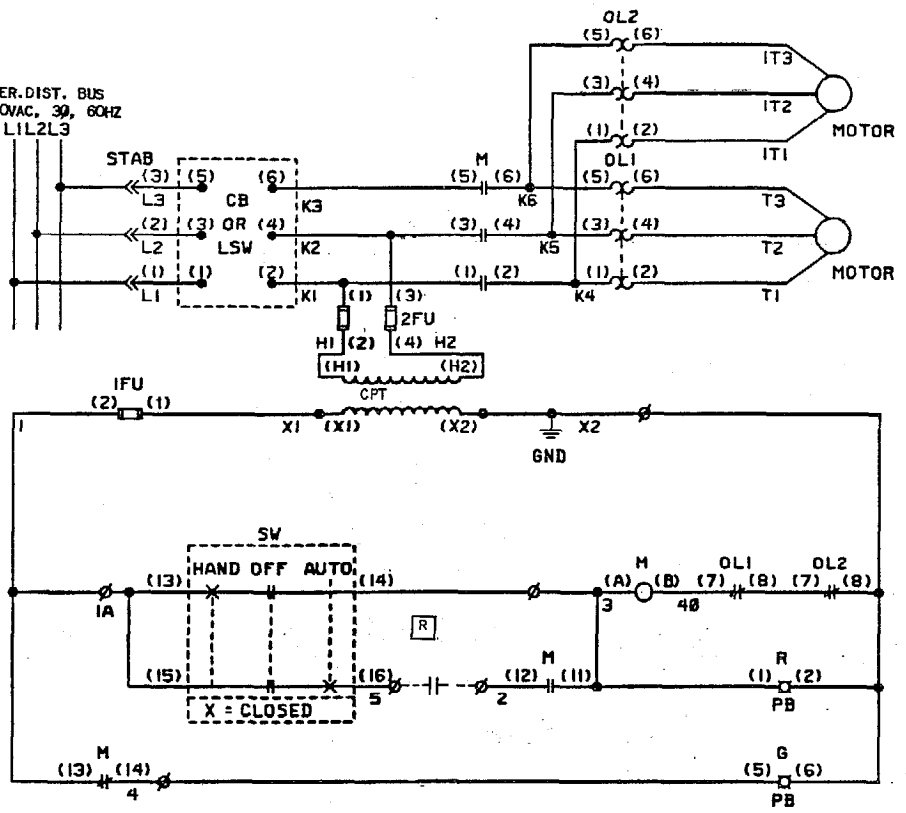
UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
EIN	NO.1 SALT WATER COOLING PUMP-EP16	150	5	TBC4 400	9.55A

R REMOTE DEVICE, SUPPLIED BY CUSTOMER

5A-2CC21

REV.3	REV.2	REV.1	ENGINEER AWJ	GENERAL ELECTRIC COMPANY DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM NO. 1 SALT WATER COOLING PUMP
REV.4	ISSUED 6-19-54	PRINTS TO	FIRST MADE FOR REGN. 420-53194-C01		229B1592 CONT. ON SH. 28P SH. NO. 28N
REV.5	MADE BY R. JOHNSON		I.C. NO.		

SH. NO. 28N
229B1592 CONT. ON SH. 28P



UNIT	DESIGNATION	H.P.	SIZE	DISCONNECT	OL HEATERS
G1J	EMERGENCY GEN LUBE PUMP-EP64/65	2.5	1	TEDL Ø07	4.19A (3/MOTOR)

-A-2HC21

REV.3	REV.2	REV.1	ENGINEER	GENERAL ELECTRIC COMPANY	ELEMENTARY DIAGRAM EMER. GEN. LUBE PUMPS
REV.4	ISSUED 6-19-84	PRINTS TO	FIRST MADE FOR REGN. 420-53194-C01	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	229B1592 CONT. ON SH. 28Q
REV.5	MADE BY R. JOHNSON		I.C. NO.		SH. NO. 28P

Table 3-2. DMCA LED Indications for Self-test Failures

CONDITION	IMOK LIGHT (CR2, Yellow)	TEST LIGHT, CR5, Red
Restart—Product code gets control of CPU	Turns on	Turns on for ¼ second
Self-tests complete	Turns off, then turns back on and stays on	Turns on for ¼ second at the end of each test
Self-test fails		Blinks rapidly until the system is 'continued' or 'restarted'

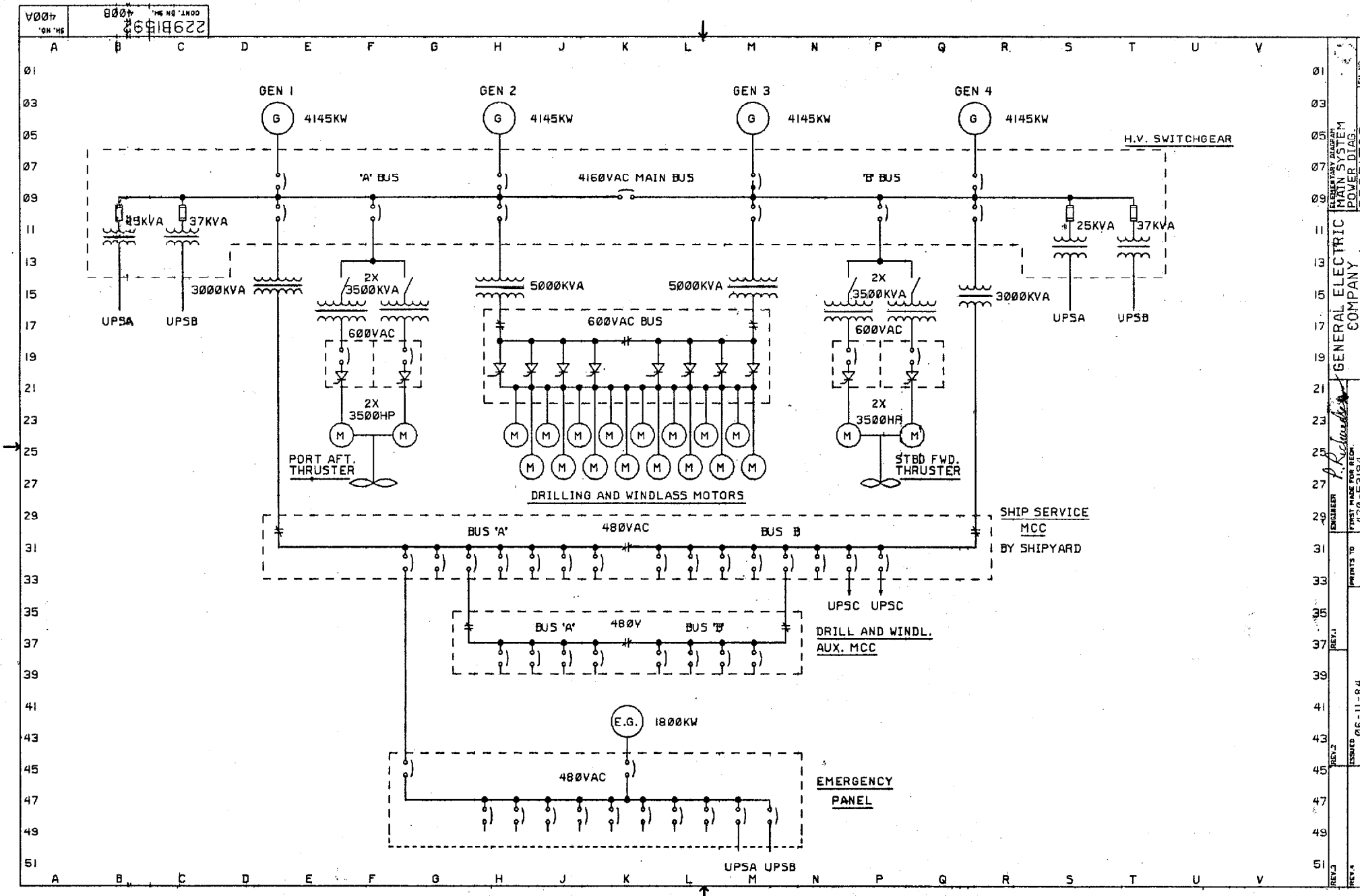
3-5. BUS TIME-OUTS

The DMC continually checks for the bus time-outs, which occur when the L-bus does not respond within 80 microseconds after an access. If the bus times out during any command, the DMC displays the following message:

=== BUS HOG ===
CS: IP xxxx: YYYY

The "xxxx" represents the current mode segment register at the time of the error, and "YYYY" represents the current offset within the segment.

For further help with DMC commands, see GEK-83839, the Distributed Micro Controller User's Manual.


GEN 1
GEN 2
GEN 3
GEN 4
4145KW
H.V. SWITCHGEAR
'A' BUS
4160VAC MAIN BUS
'B' BUS
25KVA
37KVA
3000KVA
2X 3500KVA
5000KVA
600VAC
600VAC BUS
2X 3500HP
PORT AFT. THRUSTER
STBD FWD. THRUSTER
DRILLING AND WINDLASS MOTORS
UPSA
UPSB
SHIP SERVICE MCC BY SHIPYARD
BUS 'A'
480VAC
BUS B
UPSC
DRILL AND WINDL. AUX. MCC
480V
E.G.
1800KW
EMERGENCY PANEL
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, MA. U.S.A.
ELEMENTARY DIAGRAM
MAIN SYSTEM POWER DIAG.
229B1592
SH. NO. 1
4008
CONT. ON SH.
ENGINEER
P. Richardson
PRINTS TO
FIRST MADE FOR REORD
420-5394
I.C. NO
REV.1
REV.2
REV.3
REV.4
REV.5
ISSUED
06-11-84
MADE BY
B.COBB

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC